BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 100 · Lesson 100 · GPUs: Parallel Work Beyond Graphics

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which processor is designed for throughput on massive parallel workloads?

Q2. What is parallelism?

Q3. What is a data dependency?

Q4. An image array has 4 rows × 3 columns. How many pixels?

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I Do · Read new knowledge

Key vocabulary

GPU — Graphics Processing Unit.

SIMT — Single instruction, multiple threads execution model.

throughput — Total work per time.

divergence — Threads follow differing control paths.

New learning

Observe — GPUs provide many execution units designed for high-throughput parallel operations on large batches of data.

Apply — Besides graphics, GPUs can accelerate machine learning, physical simulations, image processing and some scientific calculations.

Evaluate — Transfers between memory/device, limited parallelism and divergent branching can outweigh GPU speedup on small or sequential tasks.

Worked example

Explain why processing one million independent pixels may suit a GPU.

1. Follow the reasoning — Each pixel can receive similar operation in parallel; many GPU threads process different positions.

We Do · Practise with guidance

A sequential data-dependent algorithm calculates next item using previous result. Does GPU automatically accelerate it?

Follow the reasoning

Not necessarily: dependence limits simultaneous work, and data-transfer overhead may dominate.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Explain two features that make a GPU useful for applying same filter to millions of pixels. [6]

Question 2

Give two non-graphics GPU applications and explain how parallelism helps. [6]

Question 3

Compare CPU and GPU for 12-item sequential task versus million-element independent matrix operations. [6]

Exit Ticket · Answer before checking

A GPU is used exclusively for drawing computer-game graphics.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. GPU.

Q2. Actually running independent work at same time.

Q3. A task needs another task's result.

Q4. 12.

You Do pages · Independent questions

Q1. Many parallel lanes/threads [2], pixels independent [2], throughput on similar operations [2].

Q2. AI tensor/matrix workloads [3], scientific modelling independent calculations [3] or valid alternatives.

Q3. Short sequential: CPU avoids transfer/parallel overhead [2], massive independent: GPU throughput [2], branches/memory/bandwidth caveats [2].

Exit Ticket · Model answer

False. GPUs also accelerate AI and scientific computation.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: